HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Python** scripts for automating data extraction and transformation processes, reducing manual effort by approximately **30%**.
- Designed and implemented **Java Spring Boot** REST APIs to facilitate data exchange between internal microservices, ensuring robust error handling and efficient request processing.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, performing root cause analysis and coordinating fixes to meet strict SLAs.
- Contributed to the development of software tools for design automation within a **Python**-centric environment, focusing on enhancing RTL to GDSII flows.
- Collaborated with a geographically distributed team to deliver project goals on time, participating in code reviews and providing constructive feedback to ensure code quality.
- Provided tool support and debugging services to internal design teams, resolving issues in various phases of development to ensure smooth project execution.
- Containerized applications using **Docker** and deployed them to **AWS EC2** instances, managing configurations and basic resource allocation.

SKILLS

Languages – Python, Java, C++, SQL, R, Verilog, System Verilog

Frameworks/Libraries – Spring Boot, scikit-learn, pandas, NumPy, Flask, FastAPI

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Kubernetes

Databases – PostgreSQL, MySQL, Oracle

Tools – Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau, RStudio

Methodologies – Agile, Scrum

PROJECTS

Automated Design Flow Enhancements

- Developed a suite of **Python** scripts to automate aspects of the RTL to GDSII flow, integrating with **Git** for version control and improving efficiency for design engineers.
- Implemented unit tests for automation scripts using **pytest**, ensuring reliability and maintainability of the developed tooling.
- Provided ad-hoc debugging and support for the automation scripts, resolving ~15 issues per month reported by the design teams.

Core Java Service Development

- Engineered a **Java Spring Boot** microservice to manage design metadata, exposing **REST APIs** for CRUD operations and integrating with a **PostgreSQL** database.
- Implemented data persistence logic using **Hibernate**, ensuring efficient object-relational mapping and data integrity for ~50,000 design records.
- Wrote comprehensive **JUnit** tests and performed code reviews to maintain high code quality and adherence to design patterns for the service.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)